

1. PROJECT OVERVIEW

SmartClinic is an integrated healthcare appointment scheduling and clinical consultation workflow platform designed for multi-specialty outpatient clinics and private medical practices. The primary objective of the system is to modernize patient intake, streamline physician calendars, facilitate tele-consultations, and optimize examination room utilization across clinical departments. By replacing fragmented manual booking mechanisms with an intelligent digital platform, SmartClinic aims to improve patient engagement and maximize clinical throughput.

2. PROBLEM STATEMENT

Presently, outpatient medical centers suffer from high patient no-show rates (often exceeding 25%), severe administrative overhead caused by manual telephone bookings, and frequent double-booking of medical specialists. Patients face frustrating wait times in crowded physical waiting areas without real-time updates regarding doctor delays. Furthermore, attending physicians lack centralized pre-consultation visibility into medical intake histories, allergy alerts, and previous visit records, which hampers diagnostic efficiency and degrades the overall quality of patient care.

3. SYSTEM USERS AND ACTORS

The SmartClinic platform serves four distinct user classes:

? Patient: An individual seeking outpatient medical consultation. Patients register accounts, search for available medical specialists, reserve or reschedule appointments, submit pre-consultation intake questionnaires, and receive automated appointment alerts.

? Clinic Physician / Specialist: Medical practitioners who configure their consultation schedules, define appointment duration slots, view upcoming patient rosters, review completed intake questionnaires, and record digital clinical encounter notes.

? Front Desk Receptionist: Administrative staff responsible for checking in arriving patients, managing walk-in queues, handling manual schedule adjustments, and coordinating examination room assignments.

? Clinic Administrator: Practice operations manager responsible for physician onboarding, specialty department setup, operational hours definition, billing codes, and system audit trail inspection.

4. FUNCTIONAL REQUIREMENTS

The system shall deliver the following core functional capabilities:

? FR-01 (Patient Registration and Identity Verification): The system shall allow new patients to create an account using their national healthcare identifier or verified mobile number. Identity verification must be conducted via a 6-digit one-time password (OTP) sent via SMS.

? FR-02 (Physician Directory and Specialty Search): The system shall provide a search interface allowing patients to filter doctors by medical specialty (e.g., Cardiology, Dermatology, Pediatrics), accepted insurance provider, consultation language, clinic branch location, and earliest available consultation slot.

? FR-03 (Real-Time Appointment Scheduling and Slot Locking): Patients shall be able to view an interactive calendar displaying available consultation slots for a chosen physician. When a patient initiates a booking for a specific slot, the system must temporarily lock that slot for 10 minutes to prevent concurrent booking conflicts.

? FR-04 (Automated Multi-Channel Notifications): The system shall transmit automated appointment confirmations and reminder alerts via SMS and email to the patient 24 hours and 2 hours prior to the scheduled consultation.

? FR-05 (Digital Medical Intake and Consent Forms): Patients must be prompted to complete an electronic medical intake questionnaire (symptoms, current medications, known drug allergies) and digitally sign legal treatment consent agreements prior to consultation start.

? FR-06 (Walk-In Queue and On-Site Patient Check-In): Front desk receptionists shall have a dedicated portal to check in patients upon physical arrival at the clinic. The system must update the patient queue status in real time (e.g., Arrived, In Triage, With Doctor, Completed).

? FR-07 (Telehealth Video Consultation): For virtual consultation appointments, the system shall generate a secure, browser-based video meeting link accessible by both patient and doctor at the scheduled appointment time without requiring third-party software installation.

? FR-08 (Clinical Notes and Digital Prescription Recording): Attending doctors shall be able to record clinical diagnosis notes, ICD-10 diagnostic codes, and generate electronic prescriptions directly within the consultation encounter view upon appointment conclusion.

5. INTENTIONALLY AMBIGUOUS & BORDERLINE REQUIREMENTS (FOR AMBIGUITY DETECTION)

The following requirements contain deliberate specification gaps for analysis:

? AMB-01 (Flexible Cancellation and Immediate Refund): Patients can cancel appointments at any time if they are unhappy or change their mind, and their payment will be refunded quickly without questions asked.

(Ambiguity: Does at any time allow cancellations 5 minutes before appointment? What is the refund processing window for quickly? Is there a cancellation penalty window?).

? AMB-02 (System Responsiveness and Spike Handling): The booking interface must respond very fast and handle huge spikes in appointment bookings effortlessly. (Ambiguity: very fast and huge spikes lack concrete quantitative latency and throughput metrics, such as sub-200ms latency at 2,000 concurrent requests).

? AMB-03 (Usability for Senior Citizens): The entire web and mobile portal should be super user-friendly, highly intuitive, and easy for elderly patients to navigate without training. (Ambiguity: super user-friendly and intuitive are subjective buzzwords with no testable usability benchmarks or WCAG accessibility targets).

? AMB-04 (Medical Record Retention): All patient encounter histories, clinical notes, and video recordings must be stored safely for a reasonable period. (Ambiguity: reasonable period is legally ambiguous under medical retention mandates like HIPAA which require explicit duration such as 7 years).

6. NON-FUNCTIONAL REQUIREMENTS

The system must adhere to stringent quality and regulatory attributes:

? NFR-01 (Security and Healthcare Privacy): All sensitive Protected Health Information (PHI) must be encrypted both at rest using AES-256 encryption and in transit using TLS 1.3. Role-Based Access Control (RBAC) must strictly enforce that only designated care team members can access patient clinical files.

? NFR-02 (Availability and Reliability): The core appointment booking and clinic management services shall maintain 99.9% uptime during standard operating hours (6:00 AM to 10:00 PM local time), with scheduled maintenance windows restricted to off-peak overnight hours.

? NFR-03 (Auditability and Immutable Logging): Every access, modification, export, or deletion of patient medical records must generate an immutable, tamper-evident audit log entry containing the actor user ID, timestamp, IP address, and operation performed.

? NFR-04 (Performance and Scalability): The system must support at least 50 simultaneous clinic branches and sustain 5,000 daily booked appointments with an average 95th percentile API response time under 300 milliseconds.

7. BUSINESS RULES

? BR-01: A patient cannot have more than two active upcoming appointments scheduled within the same medical specialty on the same calendar day.

? BR-02: Physicians cannot be scheduled for continuous consultation blocks exceeding 4 hours without an automated mandatory 30-minute rest buffer.

? BR-03: Appointments where the patient has not arrived or checked in within 15 minutes of the scheduled start time shall be automatically designated as No-Show and released for emergency walk-ins.

8. TECHNICAL CONSTRAINTS & EXTERNAL INTEGRATIONS

? The system must provide standard RESTful OpenAPI endpoints for interoperability with external Electronic Health Record (EHR) systems conforming to the HL7 FHIR (Fast Healthcare Interoperability Resources) standard.

? The web application must be fully responsive across mobile, tablet, and desktop devices supporting modern evergreen browsers (Google Chrome, Mozilla Firefox, Apple Safari, Microsoft Edge).